

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester IV Examination, 2025-26 (NEP)

COMPUTER SCIENCE**Paper Code: CSMJ41 : Mathematics for Computer Science**

Time: 2:30 Hours

Maximum Marks: 70

*Instructions: Answer **FIVE** questions including compulsory Question No. 1. The figures in the right margin indicate marks.*

Question 1. (Compulsory)

- (a) Define a set and give another representation of the following sets: [4]
- (i) $\{x \mid x \text{ is an integer and } 5 \leq x \leq 12\}$
 - (ii) $\{2, 4, 8, \dots\}$
 - (iii) All the countries of the world.
 - (iv) $\{x \mid x \text{ is an integer and } x^2 = 2\}$
- (b) Discuss the limitation of propositional calculus, and how it is overcome in predicate calculus. [4]
- (c) Discuss graph coloring and give the chromatic number of a bipartite graph. [4]
- (d) Give Kolmogorov axiomatic definition of probability. [2]

Question 2.

- (a) Define number theoretic, partial, and total function. [3]
- (b) Define difference and symmetric difference of sets. Explain in general how these two set operations are achieved when sets are represented using bit strings. [4]
- (c) Suppose the domain of the propositional function $P(x)$ consists of the finite set $\{x_1, x_2, x_3, \dots, x_n\}$. Express the following statements without using quantifiers: [2]

$$\exists x P(x), \quad \forall x P(x), \quad \text{eg } \exists x P(x), \quad \forall x \neg P(x)$$

- (d) For which values of n are the following graphs planar: [2]
- (i) K_n
 - (ii) W_n
 - (iii) $K_{m,n}$
 - (iv) Q_n

- (e) Probability of the events A, B , and $A \cap B$ are 0.6, 0.3, and 0.2 respectively. Find the probability of event $arA \cap arB$. [3]

Question 3.

- (a) Define partial order relation, POSET, and lattice. [4]
- (b) Let $X = \{1, 2, 3, 4, 6, 8, 12, 24\}$ and the relation $\leq$ be such that $x \leq y$ if x divides y . Answer the following questions for the POSET $(X, \leq)$: [10]
- (i) List all ordered pairs of relation $\leq$.
 - (ii) Give relation matrix of the relation $\leq$.
 - (iii) Find the matrix of inverse relation of $\leq$ from matrix of relation $\leq$.
 - (iv) Draw the Hasse diagram of $(X, \leq)$.
 - (v) Find the maximal and minimal element, if it exists.
 - (vi) Find the greatest and least element, if exist.
 - (vii) Find all upper bounds of $\{2, 6\}$.
 - (viii) Find the least upper bound of $\{2, 6\}$, if it exists.
 - (ix) Find all lower bounds of $\{4, 6\}$.
 - (x) Find the greatest lower bound of $\{4, 6\}$, if it exists.

Question 4.

- (a) Write the negation of the following statement: [2]

$$\forall x \forall y ((x > 0) \wedge (y < 0) \rightarrow (xy < 0))$$

where the domain for both variables consists of all real numbers.

- (b) Show that the conclusion “If I do not finish writing the program, then I will wake up feeling refreshed” logically follows from the premises: “If you send E-mail message, then I will finish writing the program,” “If you do not send me an E-mail message, then I will go to sleep early,” and “If I go to sleep early, then I will wake up feeling refreshed.” [5]
- (c) Write the negation of the premises and conclusion given in 4(a). [4]
- (d) Show that $(PoQ) \iff egP \vee Q$. [3]

Question 5.

- (a) Define partition of a set. Express probability of any event in terms of the probability of the partition of the sample space and conditional probability of event with respect to the partition (Give detailed proof). [7]

- (b) Suppose a random variable X has a probability density function (pdf) given by: [7]

$$f_X(x) = \begin{cases} k(x-3)(1-x), & 1 \leq x \leq 3 \\ 0, & \text{elsewhere} \end{cases}$$

- (i) Find the value of k that makes this function a pdf.
- (ii) Find mean and variance of X .
- (iii) Find the probability of events $(-1 \leq X \leq 1)$, $(1 \leq X \leq 3)$, and $(0 \leq X < \infty)$.

Question 6.

- (a) Define an Euler circuit in a graph. Give necessary and sufficient condition that a connected graph has an Euler circuit. [3]
- (b) Can five houses be connected to two utilities without connections crossing? Justify your answer with proper reasoning. [3]
- (c) Using principle of proof by contradiction, show that conclusion Q logically follows from the premises $P, P \circ Q$. [4]
- (d) Indicate the variables that are free and bound in the following statements. Also show the scope of the quantifiers: [4]
 - (i) $(\forall x)P(x, y)$
 - (ii) $(\forall x)(P(x) \circ Q(x))$
 - (iii) $(\forall x)(P(x) \circ (\exists y)R(x, y))$
 - (iv) $(\forall x)(P(x) \circ R(x)) \vee (\forall x)(P(x) \circ Q(x))$

Question 7.

- (a) Prove that in a graph the number of nodes with odd degree must be even. [5]
- (b) A connected planar graph has 30 edges. If planar representation of this graph divides the plane into 20 regions, how many vertices does this graph have? [3]
- (c) For which values of n are these graphs bipartite? And also find their chromatic number: [4]
 - (i) K_n
 - (ii) C_n
 - (iii) $K_{m,n}$
 - (iv) Q_n
- (d) What will be the maximum chromatic number of Hasse diagram of a POSET? Justify your answer with proper reasoning. [2]

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